CHT PLATFORM · DATA GOVERNANCE

Data Flow Diagram

Every flow that moves data between Zoom, learners, third-party platforms, Platform, Content Hub, Reports and Companion. For each flow: source, destination, data, frequency, transformation, owner and what happens when it fails.

How to read this document

Section 2 is the diagram, with each arrow labelled by a flow ID. Section 3 is the flow inventory in the requested columns, using the same IDs. Section 4 shows what is switched on in dev versus prod. Section 5 groups the failure behaviours. Section 6 lists the gaps to fix. Companion document: *CHT Data Dictionary & Metric Definitions*.

Version	1.0 (draft for engineering and product review)
Scope	`develop` branch of each repo plus the Terraform variables for dev and prod; read-only static review
Repos	`cht-platform-tool` @ `ad32e73` · `cht-content-hub` @ `72d245d` · `cht-reports` @ `a641074` · `cht-companion` @ `b9a0e7e`
Method	Each failure behaviour was taken from the code path that handles it (file references in grey). Third-party retry policies (Zoom, Stripe, Bill.com, AWS Lambda) are marked *inferred*.
Classification	Confidential. Internal use only.

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1 Summary

The system has 35 flows in six groups: capture into Platform, Platform outbound, Content Hub ingestion, the report pipeline, Companion, and delivery. The report pipeline is the most robust: it uses SQS with three attempts, a dead-letter queue, a timeout sweeper and idempotent S3 keys. The weakest flows are the ones that feed it. Several of them are fire-and-forget, manual, or switched off in production.

PROD HUB HAS NO SCHEDULED REFRESH

Content Hub's weekday export ingest (`platform_export_ingest`) and S3 transcript ingest (`vtt_object_ingest`) run in dev only. In prod, Hub refreshes Platform data only when Platform triggers it: on campaign link, first transcript pull, Generate, or Refresh data.

TRANSCRIPTS DEPEND ON A MANUAL PULL

Zoom recordings and transcripts reach S3 only when an admin clicks Sync or Pull. Reports wait up to 180 minutes for a recent session's transcript, then fail with a "transcript did not arrive" message.

FIRE-AND-FORGET FLOWS CAN LOSE DATA SILENTLY

HubSpot contact sync, the Hub HCP roster sync and the background export-ingest triggers only log warnings on failure, with no retry. HubSpot sync even reports success when it fails.

SOME WEBHOOKS ACKNOWLEDGE FAILURES

The Jotform webhook returns 200 when processing fails, so Jotform never retries. The Zoom webhook returns 200 on an invalid signature and skips verification when the signature header is missing.

QUEUES ARE PROTECTED BUT NOT ALL MONITORED

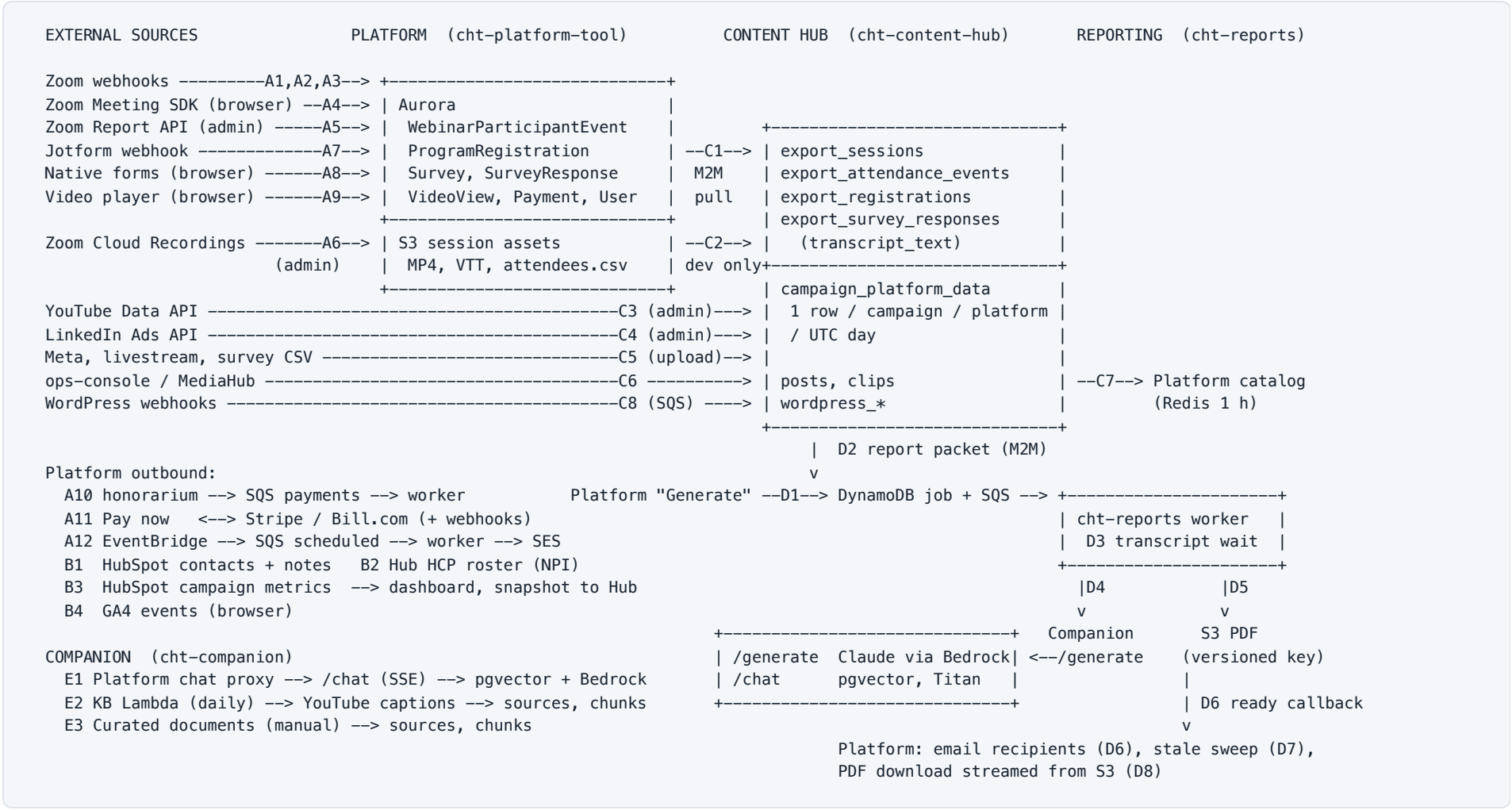
Platform's payment, email and scheduled-job queues have dead-letter alarms. The cht-reports request queue has a dead-letter queue but no alarm. Platform's stale-report sweeper is the only signal for a stuck report.

OWNERSHIP IS CLEAR EXCEPT AT THE EDGES

Each repo owns its own flows. Post statistics (ops-console and MediaHub) and the Zoom, YouTube, LinkedIn and HubSpot APIs are external, so their freshness and retry behaviour are outside our control.

2 Diagram

Arrows are labelled with the flow ID from Section 3. Labels in brackets mark manual or dev-only flows.



3 Flow inventory

Retries failures are retried automatically; Logged only failure is logged and the data may be stale or missing; Silent loss data can be lost with no retry and no visible error. File references are relative to each repo's backend source root.

ID	Description	Source	Destination	Data	Frequency	Transformation	Owner	Failure behavior
A. CAPTURE INTO PLATFORM								
A1	**Zoom participant events**	Zoom webhook `*.participant_joined` and `*.participant_left`	Platform `WebinarParticipantEvent`	Participant email, name, Zoom ID, join or leave time	Real time (event)	Email lower-cased and matched to a CHT user; one JOINED or LEFT row; no duration stored. Email and name are also pushed to HubSpot (B1).	Platform	Logged only Invalid signature: 200 and dropped. Missing signature header: accepted without verification. Unknown meeting: logged and dropped. Database error: 5xx, and Zoom retries (inferred). platform `webinars/zoom-webhook.service.ts:117–123, 333–426`
A2	**Session ended and auto-verify**	Zoom webhook `meeting.ended`, `webinar.ended`	`Program.zoomSessionEndedAt`; registration attendance status	Actual end time	Real time	End time only moves later. Registrants whose email appears on a join row become VERIFIED; then enrollment and survey email.	Platform	Logged only Auto-verify errors are logged and the end time is kept. Registrants stay PENDING_VERIFICATION until an admin acts or an import re-runs auto-verify. platform `zoom-webhook.service.ts:205–226`
A3	**Webinar auto-import**	Zoom webhook `webinar.created`	`Program` (DRAFT)	Topic, start, duration, meeting ID	Real time	Creates a draft program with credits = 0. `meeting.created` is ignored.	Platform	Logged only No program is created; an admin creates it by hand.
A4	**In-app join**	Zoom Meeting SDK in the browser	`WebinarParticipantEvent`	Signed-in user's email; JOINED or LEFT	On join or leave	Stored as source WEBHOOK (the `MEETING_SDK` value is never used), tagged in the raw payload.	Platform	Logged only A failed client call writes no row; attendance then relies on A1.
A5	**Attendee report import**	Zoom Report API `/report/webinars|meetings/{id}/participants`	`WebinarParticipantEvent` (REPORT_IMPORT); S3 `attendees.csv`	Per-segment join, leave and duration; host flag	Manual (admin, single session or bulk up to 12 months)	Upsert on (program, participant, event, source, joinTime). Unlinked sessions are staged. Optional auto-verify (off by default).	Platform	Retries A 404 is retried with the meeting UUID. Job state and errors are kept in `ZoomAttendanceImportJob`. The lock is held in memory, so two tasks could run the same import (inferred). platform `zoom-recordings/zoom-attendance-import.service.ts`
A6	**Recordings and transcripts**	Zoom Cloud Recordings API	S3 session assets `zoom-recordings/…`	MP4, transcript VTT, closed captions	Manual (admin Sync, then Pull)	Files copied to S3. The first transcript for a linked program triggers C1 (`platform_zoom`).	Platform	Logged only Each file is marked FAILED; an admin pulls again. With no automatic pull, reports wait for the transcript (D3) and then fail.

ID	Description	Source	Destination	Data	Frequency	Transformation	Owner	Failure behavior
A7	**Jotform submissions**	Jotform webhook	`SurveyResponse`; registration acknowledgement	Answers, submission ID, hidden `user_id`	Real time	Deduplicated on submission ID. FEEDBACK keeps the first submission; other types keep the latest. May create a survey bonus payment.	Platform	Silent loss A missing `user_id` means the submission is dropped. Processing errors return 200 with an error field, so Jotform does not retry. platform `jotform/jotform-webhook.controller.ts:50-67`
A8	**Native survey submit**	Learner browser	`SurveyResponse`	Answers, client score	On submit	Create or update; side effects: HubSpot contact (B1), survey bonus, FEEDBACK acknowledgement.	Platform	Retries The error is shown to the learner, who resubmits. Side effects are best-effort.
A9	**Video progress**	Learner video player	`VideoView`, `ProgramEnrollment`	Watched seconds, progress, completed	Every 15 s, on completion, on tab hide	Program progress = mean of video progress. Complete at 95%. Non-live completion queues an honorarium.	Platform	Retries Progress is held in browser localStorage and sent again on the next sync.
A10	**Honorarium request**	Platform API	SQS payments queue, then worker, then `Payment`	User, program, amount, idempotency key	On learner request	Worker inserts a PENDING row; a duplicate key is ignored.	Platform	Retries Transient errors are retried up to 3 times, then go to the dead-letter queue (alarm configured). Invalid messages are deleted immediately. platform `worker/utils/sqs_base.py`; `tf/modules/messaging/sqs`
A11	**Payouts**	Admin "Pay now"; Stripe and Bill.com webhooks	Stripe or Bill.com; `Payment` status; `User.totalEarnings`	Amount, rail references, status	On demand, plus provider webhooks	PENDING, then PROCESSING, then PAID or FAILED. Earnings counter incremented on PAID.	Platform	Logged only A provider error marks the payment FAILED with a reason, and an admin retries. Webhook ordering can double count or drift the earnings counter.
A12	**Session reminders**	EventBridge rule	SQS scheduled-jobs queue, then worker, then SES email	Approved registrations starting in 18 to 30 hours	Every 6 h in prod, 12 h in dev	One email per registration; `reminder24hSentAt` is stamped.	Platform	Retries 3 attempts, then the dead-letter queue (alarm configured). The stamp prevents duplicate emails.
B. PLATFORM OUTBOUND								
B1	**HubSpot contact sync**	Platform	HubSpot CRM (contacts, notes)	Email, name, specialty, institution, NPI; registration events	On signup, profile update, survey, registration, Zoom join	Batch upsert; custom registration properties; timeline note.	Platform	Silent loss Fire-and-forget. Errors are swallowed and the result still reports success. No retry and no backfill. platform `outbound-sync/outbound-sync.service.ts:60-81`
B2	**Hub HCP roster**	Platform	Content Hub HCP roster	HCP profile keyed by NPI	Same triggers as B1	Sent only when the NPI has 10 digits.	Platform	Silent loss Fire-and-forget. Users without an NPI never reach Hub.

ID	Description	Source	Destination	Data	Frequency	Transformation	Owner	Failure behavior
B3	**HubSpot campaign metrics**	HubSpot Marketing API	Platform campaign dashboard and funnel; Hub `campaigns.hubspot_raw_data`	Sessions, contacts, email, landing pages, forms, social	On demand (page load or admin sync)	Recursive key matching into normalised totals; the raw snapshot is stored in Hub.	Platform	**Logged only** Falls back to the cached Hub snapshot; the response shows whether data is live, cached or list-only.
B4	**Web analytics**	Learner browser	GTM `GTM-MR68PMPZ`, GA4 `G-EXVD5CJLQL`	Video start, progress, complete; clip view, share	Real time	None (client events).	Platform	**Silent loss** Blocked by ad blockers and consent settings. Not stored by CHT.
C. CONTENT HUB INGESTION								
C1	**Platform export ingest**	Platform `GET /api/export/reports/campaigns/:id/input-packet` (M2M)	Hub `export_sessions`, `export_attendance_events`, `export_registrations`, `export_survey_responses`	Sessions, transcript keys, rolled-up attendance, registrations, surveys	Campaign link, first transcript, Generate, Refresh data. Weekday 14:00 UTC schedule: **dev only**.	Platform rolls up attendance per person (report data preferred) with no date window. Hub upserts by dedupe key and never deletes.	Platform (packet), Content Hub (ingest)	**Logged only** Background triggers only log a warning. Generate continues with stale data (`warehouseSync=failed`). Refresh data returns the error. Hub returns 502 or 503 and records `export_ingest_runs`. Re-linking a program leaves orphaned rows in the old campaign. platform `content-hub/campaign-export-ingest.service.ts`; `reports/reports.service.ts:596-608`
C2	**Transcript text**	S3 `ObjectCreated` on `zoom-recordings/*.vtt`	Hub `export_sessions.transcript_text`	VTT transcript	Event. **Dev only** (prod notification not wired).	VTT parsed to text; skipped if unlinked, no session, no campaign, or the hash is unchanged.	Content Hub (Lambda), Platform (bucket notify)	**Logged only** Skip statuses are logged. Lambda async retries apply (inferred). In prod, the transcript reaches Hub only through C1.
C3	**YouTube metrics**	YouTube Data API v3	Hub `campaign_platform_data`	Video ID, title, published date, views, duration	On demand (admin Sync or Sync all)	Channel uploads filtered to the reporting period; one row per campaign per UTC day (same-day sync overwrites).	Content Hub	**Logged only** The connector error is saved on the sync run and returned as 502. The previous snapshot remains. Missing configuration returns 400. hub `services/platform_data.py:338-450`
C4	**LinkedIn metrics**	LinkedIn Ads API `adAnalytics` (202501)	Hub `campaign_platform_data`	Impressions, clicks, video views, completions, dwell	On demand	Daily rows per campaign; CTR and completion rate computed.	Content Hub	**Logged only** Same as C3. An expired token fails the sync until it is refreshed.
C5	**Uploaded platform data**	Admin CSV upload (Meta, livestream, survey platforms)	Hub `campaign_platform_data` (source=upload)	Rows as exported by the platform	Manual	Parsed rows stored as JSON; no metric math.	Content Hub	**Logged only** Parse errors are returned to the uploader. Nothing is collected unless someone uploads.

ID	Description	Source	Destination	Data	Frequency	Transformation	Owner	Failure behavior
C6	**Post statistics**	ops-console and MediaHub	Hub `posts`	Views, likes, comments, shares, impressions	External schedule (not in these repos)	Written into `posts` by the external system; Hub sums per clip on read.	External (ops-console)	**Logged only** Not visible to these repos. `stats_synced_at` shows the age of each post.
C7	**Public catalog**	Hub `/clips`, `/doctors`, `/wordpress/*`	Platform catalog API and pages	Clips with summed counts, doctors, WordPress posts	Per request; Redis cache 1 h (clips), 5 min (WordPress)	Passed through; total from `X-Total-Count`.	Content Hub (API), Platform (cache)	**Logged only** Cached data is served until it expires. On a cache miss, Hub errors reach the page; WordPress 401 or 404 returns an empty list.
C8	**WordPress mirror**	WordPress webhooks	SQS, then Hub `wordpress_*`	Posts, series, categories, tags	Event (prod and dev). Nightly drift reconcile: dev only.	Projected into relational tables.	Content Hub	**Retries** SQS redelivery. Without the nightly reconcile in prod, a missed event stays missing.
D. REPORT PIPELINE								
D1	**Report request**	Platform "Generate" or "Regenerate"	DynamoDB report job; SQS report-requests queue	Campaign, template, frozen window, sources	On demand	Lock per campaign and template; C1 runs first (best-effort); message enqueued.	Platform	**Retries** If the enqueue fails, the job is marked failed, the lock is released, and the admin sees 503. platform `reports/reports.service.ts:137–151`
D2	**Report packet**	Hub report-packet API (M2M)	cht-reports worker	Sessions, transcripts, attendance summary, surveys, platform data, HubSpot, KOLs, template, completeness	Per run	Hub computes registered and attended counts and average minutes; nameless attendee list.	Content Hub	**Retries** Any error fails the attempt; SQS redelivers (maximum 3), then the dead-letter queue. **No dead-letter alarm.**
D3	**Transcript wait**	cht-reports	SQS (re-queue with delay)	Request ID	Every 300 s, up to 180 min (sessions in the last 24 h)	Status `waiting_for_transcript`.	Reports	**Logged only** After 180 minutes the job is marked failed with "Zoom's transcript did not arrive"; the admin generates again later. reports `reports/orchestrator.ts:96–118`
D4	**Narrative generation**	cht-reports	Companion `/generate`, then Amazon Bedrock (Claude Sonnet)	System prompt, cleaned transcript (100k characters), packet context	Per run	13 narrative sections returned as JSON; output limited to 16k tokens.	Reports (caller), Companion (service)	**Retries** Bedrock: 270 s read timeout, 2 attempts, then 502. Rate limit returns 429. The whole request is retried from the start (maximum 3). A cut-off output fails the attempt rather than being published.
D5	**PDF upload**	cht-reports	S3 reports bucket; DynamoDB status complete	Rendered PDF	Per run	HTML rendered to PDF; key versioned by edit attempt (v1, v2…).	Reports	**Retries** The key is deterministic, so a retry overwrites the same version. An already-complete job is skipped on redelivery.

ID	Description	Source	Destination	Data	Frequency	Transformation	Owner	Failure behavior
D6	**Report ready**	cht-reports	Platform `POST /api/internal/reports/:id/ready`, then SES email	Report ID, campaign, version	Per completed run	Platform claims `notified_version` once per version, then emails recipients.	Reports (call), Platform (email)	Retries 3 attempts with backoff (one token refresh on 401). Never fails the report. The conditional write prevents duplicate emails. reports `notify/report-ready.service.ts`
D7	**Stale-report sweep**	Platform	DynamoDB report job	Status, `updated_at`	On read (list or get)	No progress for `REPORTS_STALE_MINUTES`: mark failed ("Timed out") and free the lock.	Platform	Logged only This is the only signal for a crashed worker or a dead-lettered message.
D8	**Report download**	S3 reports bucket	User (through Platform)	PDF only	On demand	Streamed through Platform; the S3 URL is never exposed and non-PDF keys are rejected.	Platform	Retries 409 "not ready" until complete; the user retries.
E. COMPANION								
E1	**Chat**	Platform chat proxy (signed-in user)	Companion `/chat` (SSE), then pgvector and Bedrock; Platform conversation tables	Question, last 20 turns (up to 8,192 characters); answer tokens, citations	Per message	Retrieval k=10 with neighbouring chunks; streamed answer; conversation saved by Platform.	Platform (proxy), Companion (service)	Logged only Not configured: 503. Upstream status passed through. Rate limit: 429 (20/min, 500/day). Redis down: 503 (fails closed). Partial answers are saved.
E2	**Knowledge-base ingest**	YouTube captions (KB Lambda)	Companion `sources`, `chunks`, `ingest_jobs`	Caption text, metadata, Titan embeddings	Daily (`rate(1 day)`, dev and prod)	Chunked and embedded (1024-d); status pending until approved.	Companion	Silent loss Caption downloads currently return 403, so nothing is queued and no new content arrives.
E3	**Curated documents**	Admin upload	Companion `sources`, `chunks`	Documents	Manual	Chunked, embedded, approved.	Companion	Logged only The job is marked failed in `ingest_jobs` with an error message.

4 Dev versus prod

Several scheduled flows are switched on in dev and off in prod. Data in prod is therefore fresher only when someone acts in Platform.

Flow / job	Repo	Dev	Prod	Effect in prod
C1 scheduled export ingest	Content Hub	On weekdays 14:00 UTC	Off	Hub refreshes only on link, transcript pull, Generate or Refresh data.
C2 VTT transcript ingest	Content Hub + Platform	On	Off	Transcripts arrive through C1 only.
C8 WordPress ingest	Content Hub	On	On	None.
WordPress nightly reconcile	Content Hub	On 03:00 UTC	Off	Missed webhooks are not repaired.
Post tagging	Content Hub	On every 12 h	Off	New posts are not auto-tagged.
HCP intel, OpenAlex, KOL–HCP matcher, playlist tagger	Content Hub	Off	Off	KOL intelligence is not refreshed in either environment.
A12 session reminders	Platform	Every 12 h	Every 6 h	None.
E2 KB ingest	Companion	Daily	Daily	Blocked by caption 403s in both.
Companion scheduled scaling	Companion	Off 01:00–13:00 UTC and at weekends	Always on	In dev, chat and report generation fail outside working hours.

5 Failure-handling patterns

Pattern	Flows	Behaviour	Visibility
Queue with retry and DLQ	A10, A12, C8, D1–D5	Message is left on the queue on error, redelivered up to 3 times, then moved to the dead-letter queue. Bad messages are deleted at once.	DLQ alarms on Platform queues; **none on the reports queue**.

Pattern	Flows	Behaviour	Visibility
Bounded retry in code	A5, D4, D6	Fixed attempts with backoff (Bedrock 2 attempts, ready callback 3 attempts, Zoom 404 retried with the UUID).	Logs.
Best-effort, then continue	A2, C1 (on Generate), B3	The failure is logged and the main action continues with older data (stale warehouse, cached HubSpot).	Flags such as `warehouseSync=failed` and `dataSource`.
Fire-and-forget	B1, B2, C1 (link, transcript)	No retry and no outbox. A failure means the data is missing until the next trigger.	Warning logs only.
Acknowledge and drop	A1 (bad signature), A7	Returns 200 to the sender, so it never retries.	Logs only.
Manual recovery	A5, A6, C3–C5, D3	An admin re-runs the import, pull, sync or upload, or generates the report again.	Job and sync status tables, report `last_error`.
Fail closed	E1 rate limiter	Returns 503 when Redis is down instead of serving without limits.	Error response.
Timeout sweeper	D7	Marks stuck reports failed and frees the lock.	Report status in Platform.

6 Gaps and recommendations

#	Gap	Recommendation	Owner
1	Prod Hub has no scheduled export ingest (C1) and no VTT ingest (C2).	Set `platform_export_base_url` for prod and enable `platform_export_ingest`; wire the prod S3 notification and enable `vtt_object_ingest`.	Content Hub, Platform
2	Recordings and transcripts need a manual pull (A6), so reports can wait 3 hours and then fail.	Subscribe to Zoom `recording.completed` or `recording.transcript_completed` and pull automatically.	Platform
3	Fire-and-forget outbound sync (B1, B2, C1 triggers).	Send through an SQS outbox with retries and a DLQ, and stop reporting HubSpot success on failure.	Platform
4	Jotform webhook returns 200 on error; submissions without `user_id` are dropped (A7).	Return 5xx on transient errors; record dropped submissions for review.	Platform

#	Gap	Recommendation	Owner
5	Zoom webhook accepts requests without a signature header (A1).	Reject unsigned events when a secret is configured.	Platform
6	The reports DLQ has no alarm (D2–D5).	Add a CloudWatch alarm on `…-report-requests_dlq`, matching Platform's queue alarms.	Reports
7	Attendance and survey changes do not refresh Hub (A2, A5, A7, A8, then C1).	Trigger C1 after auto-verify, report import and survey submit, or rely on gap 1's schedule.	Platform
8	Platform data (C3, C4) refreshes only on demand.	Schedule `sync_all_platforms` daily (the `next_sync_at` field already assumes it).	Content Hub
9	Re-linking a program leaves orphaned rows in the old campaign (C1).	Delete or retarget the old campaign's rows on unlink.	Content Hub
10	Companion KB receives no new content (E2).	Fix caption access, or ingest from the Zoom VTTs already in S3.	Companion